

NATIONAL UNIVERSITY
of Computer & Emerging Sciences, Lahore

Project Phase 1 (Track C)

Memory Priority Scheduling Algorithm for Cloud Data Center Based on Machine Learning Dynamic Clustering Algorithm

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Course: Artificial Intelligence (AI-2002)

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Group ID: 3

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Problem Statement

Cloud data centers (CDCs) waste significant energy because most virtual machine (VM) scheduling algorithms optimize central processing unit (CPU) utilization while ignoring memory, leaving physical machines (PMs) with unbalanced CPU/memory usage. This forces more PMs to stay active than necessary, increasing energy consumption. The paper asks: **how can a scheduling algorithm jointly balance CPU and memory utilization to reduce the number of active PMs and lower CDC energy use?**

Research Paper

Memory Priority Scheduling Algorithm for Cloud Data Center Based on Machine Learning Dynamic Clustering Algorithm [1]

Dataset

https://github.com/alibaba/clusterdata/blob/master/cluster-trace-v2018/fetchData.sh [2]

(real-world cloud/data-center workload and resource-usage trace data)

Justification for Selection

It gives explicit pseudocode (Algorithm 1), worked numerical examples (Tables II–IX), and simple formulaic metrics (Equations 1–4), making it unusually reproducible. The machine learning (ML) component is just K-means — lightweight, fast, no graphics processing unit (GPU) or large datasets needed. Baselines (like First-Fit) are trivial to implement, so meaningful comparison is feasible in a course timeline.

Core Contribution

PMPD balances CPU and memory jointly instead of separately, via two rules:

1. **PM clustering** — group PMs by CPU-to-memory utilization ratio using K-means, updated dynamically.
2. **VM deployment** — assign each VM to the cluster whose imbalance it corrects, then place it on the PM with closest remaining runtime to minimize wasted active time.

This reduces active PM count and light-load time, improving CPU/memory utilization and cutting CDC energy consumption by ~14–20% versus FF, WS, task placement optimization (TPO), and differential evolution-enhanced regional particle swarm optimization (DE-ERPSO) in the paper's experiments.

Project Scope

1. Implement the PMPD algorithm (K-means PM clustering + CPU-to-memory ratio-based VM deployment rule) as a shared, reusable simulation core.
2. **Synthetic experiment:** Generate PMs and VMs matching the paper's parameter ranges (Table VI, Table X), and reproduce the VM-count sweep and minimum-utilization-threshold sweep experiments, computing the same four metrics (light-load time, CPU utilization, memory utilization, energy consumption).
3. **Real-data experiment:** Use the Alibaba Cluster Trace v2018 (`batch_instance` data), extract and clean a sample, derive PM/VM parameters from it, and run the same PMPD core and metrics.
4. Implement First-Fit as a baseline and run it through both experiments for direct comparison.
5. Compare synthetic results directionally and (where feasible) numerically against the paper's reported figures; compare real-data results directionally against the same baseline.
6. Discuss any differences between synthetic and real-data outcomes, and what that implies about the algorithm's robustness.

References

[1] B. Liang and D. Wu, "Memory Priority Scheduling Algorithm for Cloud Data Center Based on Machine Learning Dynamic Clustering Algorithm," in IEEE Transactions on Industrial Informatics, vol. 21, no. 4, pp. 3485-3492, April 2025, doi: 10.1109/TII.2025.3528574.

[2] Alibaba Group. (2020). *Alibaba Cluster Data* [Data set]. GitHub. github.com